Name: ______________________________ **Date:** _________________ **Class:** _______________

MyDesign Portfolio

Element L Initial Template AT - 6 points

Element L: Presentation of Designer's Recommendations

Content of Recommendations *(L1, L2)*

In K3, you provided the general lessons you learned related to engineering design. Now, you are asked to provide recommendations for continuing work on this project.

Answer these questions with **specific recommendations** based on your testing data, expert/stakeholder feedback, and any new research you've gathered. These recommendations may include process improvements (how you worked) and/or project improvements (actual design-specific decisions you made).

Provide recommendations to someone doing a similar project that might build or continue upon your work. For example: "Taking into account <something we did>, we believe doing __________ would improve _________ because ________."

- What could be done next to go further with this project?

Taking into account our testing results and feedback, we believe the next step would be to build a stronger, more stable frame for the system. Our current setup needs to be held up by hand, which is not practical. A new frame made of lightweight wood or sturdy plastic would allow the system to stand on its own. We also recommend adding a proper water container, like an IV bag or water tank, that hangs above the plants to keep a steady water flow. Testing with a moisture sensor could help check when plants actually need water, which would improve water use and plant health.

- What would you keep if you did the project again? Generate a consistently **detailed list** of what would be kept in the prototype or testing plans. (4-5 sentences.)

engineering 4 us all

Student Handout

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We would keep the IV needle and tubing system because it gave us better control over how much water each plant got. We would also keep the idea of using clear tubing so we can see if the water is flowing or if there are any blockages. The setup steps we created were easy to follow, and students were able to build it without needing help, so we would keep those instructions. We'd also stick with tracking results like water flow and how long the system runs. Our testing plan helped us find problems early, so we would definitely keep that too.

- What would you change in your current prototype or testing plan? Generate a consistently **detailed list** of HOW these details would be implemented. (4-5 sentences.)

We would change the frame so the system doesn't have to be held during testing. To do this, we'd use a light, strong material to build a stand or support that holds everything in place. We would also improve the water source by using a real IV bag or gravity-fed container instead of plastic bottles. The tubing would be sealed better to stop leaks, maybe using stronger glue or clamps. For testing, we'd run more trials (at least 5–6) to get better data and make sure the system works in different situations. Finally, we'd mark the best IV needle pressure point sothe setup is quicker.

- What would have made this project easier for you to complete? Generate a consistently **detailed list** of things that would make the project easier. (You may make a list or use 4-5 sentences.)

One thing that would have helped a lot is having a proper frame or stand ready at the beginning. Since we had to hold the system up by hand, it made testing harder and less

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accurate. If we had a stable structure already built, we could have focused more on fixing the water flow instead of holding everything in place.

Getting the correct materials from the start, like real IV bags, stronger tubing, and better connectors, would have saved us time and stress. We had to make do with plastic bottles and cheaper materials, which didn't always work well and caused leaks.

Talking to an expert in plant care or irrigation earlier would have made a big difference. They could have told us which materials and flow rates were best for keeping the plants healthy and how to prevent mold or overwatering. This would have saved us from a lot of trial and error.

We also needed more class time to test our system and make adjustments. Sometimes we ran out of time during testing, so we couldn't always fix problems right away or collect more data.

Lastly, having a simple tool to measure soil moisture, like a moisture meter, would have helped us know exactly when the plants needed water. That way, we could have tested how well the system matched the plants' needs, instead of just guessing based on how they looked.

Step L: Presentation of Designer's Recommendations						
L1. Content of recommendations. Recommendations include __________.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0

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	specific and detailed ways the project can be improved and a rationale for why they are needed	specific and detailed ways the project can be improved	some specific ways the project can be improved	one way the project can be improved	vague recommendation(s) for improvement of the project	no recommendations for improvement of the project
L2. Detailed plans for implementation of improvements. Plans for implementation of improvements are __________.	score = 5	score = 4	score = 3	score = 2	score = 1	score = 0
	consistently detailed	generally detailed	minimally detailed and may include one or more caveats	partially detailed and/or overly general	vague/unclear or minimally related to the recommendations provided	not offered